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Breast cancer: definition, epidemiology, risk factors

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Introduction

Breast cancer has a high incidence, numerous therapeutic options exist, and patients have high expectations of interdisciplinary medical teams.

Breast cancer is not a uniform disease!

- Certain breast cancers should be understood as a “systemic disease” at the time of detection due to their tendency for metastatic dissemination.
- Other breast cancers – particularly hormone-sensitive carcinomas – remain a local/regional disease for a long time. These cancers can be successfully treated surgically, even in the presence of lymph node involvement.

Stem cell theory has received increasing attention in the last few years. The theory proposes that - at first - one stem cell of the mammary gland epithelium mutates and proliferates, however, it can differentiate in several pathways to a certain degree. One pathway is to give rise to a hormone-sensitive, highly differentiated carcinoma. In the course of the disease, the metastasized stem cell of this carcinoma can, however, form a poorly differentiated carcinoma without hormone receptors. This metastasized stem cell displays a different morphology than the original breast carcinoma at the origin of the metastasis. [P10]

Definition

Healthy breasts consist of fatty tissue, glandular tissue, the surrounding skin and the nipple-areola complex.

WHO histological classification: an overview

Noninvasive breast cancer

- Ductal carcinoma in situ = DCIS
- Lobular neoplasia = LN (previously: LCIS)
- Paget's disease of the nipple (= adenocarcinoma cells “in situ” in the epithelium of the skin)

Invasive breast cancer

- Invasive ductal cancer
- Invasive lobular cancer
- Inflammatory breast cancer
- Specific types of cancer (tubular, papillary, cribriform, mucinous, medullary, etc.)
- Invasive Paget's disease (Paget cancer) of the nipple

The 2012 WHO classification provides a histological grid consisting of 18 different histological types, yet these have weak correlations with the biological and clinical characteristics of tumors. [B01]

In contrast to this “traditional” (histological) classification, the molecular biology classification, which is more meaningful for tumor biology, has been established. This classification was developed based on gene expression profiles and distinguishes luminal and basal types of breast carcinoma (table 1).

The breast cancer entities that can be distinguished with the help of gene expression profiles correspond relatively well to certain immunohistochemistry patterns. Because breast cancer gene expression profiles are not routinely determined, everyday clinical practice uses immunohistochemical determination of the estrogen receptor, progesterone receptor, HER2 receptor (or immunofluorescence of the HER2 gene), and the proliferation fraction (Ki 67 or MIBB1).

Table 1 (EGONEplus) [P06]

Gene expression type	Estrogen receptor	Progesterone receptor	HER2	CK5/6 or EGFR	Ki 67 or MIBB1
Luminal A	High	High	Negative	–	Low
Luminal B	High or weak	Weak or negative	Negative	–	High
Luminal HER2	Positive	Positive or negative	Positive	–	–
HER2 positive	Negative	Negative	Positive	–	–
Basal cell	Negative	Negative	Negative	Positive	–
Triple negative, not basal cell	Negative	Negative	Negative	Negative	–

The basal type is more common in women of African origin and has the following characteristics: it is always estrogen receptor negative, poorly differentiated, HER2-negative.
The triple-negative phenotype is frequent in patients with a BRCA1 mutation.
For this reason, the occurrence of triple-negative breast cancer before age 60 is an accepted indication for BRCA testing in Switzerland (mutation probability: > 10 %).
In accordance with European guidelines (ESMO), testing should be performed for all women < 50 years with a triple-negative carcinoma.

Non-Invasive Breast Carcinomas: Two Main Types

Ductual carcinoma in situ = DCIS: "Intraductal Carcinoma"

- Cells are degenerated inside the lining of the milk ducts.
- Usually occurs after the age of 40.
- More common than lobular neoplasia (LN). It is more frequently diagnosed due to improved breast imaging (breast MRI, improved mammography techniques, and mammography screening). Usually identifiable with mammography due to microcalcifications.
- Heterogeneous histology (subtypes). Heterogeneous differentiation (high grade vs. non high grade).
- Local precancer of invasive ductal breast carcinoma! Without treatment, 30-40% of DCIS cases become invasive carcinoma.
- Must be treated locally! Therapeutic strategy: Excision with surrounding healthy

tissue (= excision of the affected milk duct, or – if extensive – mastectomy); post-operative radiation therapy for the remaining mammary gland in the presence of increased risk of relapse.

Ductal carcinoma in situ (DCIS) is a precancer!

Lobular neoplasia = LN (previously: Lobular carcinoma in situ = LCIS)

- Cells are degenerated inside the lining of the gland lobules.
- Usually occurs in premenopausal (younger) women.
- Often an incidental finding in excised mammary gland tissue without correlating imaging.
- Rarer than DCIS. It is being discovered more often due to more frequent breast interventions.
- There are various histological subtypes.
- It is seldom detected by palpation nor by mammography.
- NOT a precancer by definition but considered a risk indicator given that in future decades LN carriers may develop invasive ductal (rarer: invasive lobular) carcinoma at the same location, in another quadrant, or in the contralateral breast. This risk is estimated to be 1 % per year.
- After a diagnosis of LN, both breasts should be routinely checked with imaging at short intervals (e.g., annually).
- Chemoprevention with tamoxifen (Nolvadex®, generic drugs) should be considered. A (bilateral) prophylactic mastectomy is not justified.

Lobular neoplasia (LN) is not precancerous! Internationally, the term LN has replaced the formerly used terms lobular carcinoma in situ, LCIS and CLIS.

On the other hand, its two variants, florid LCIS (FLCIS) and pleomorphic LCIS (PLCIS), present different morphological and genetic characteristics and they present a higher probability to be possible obligatory precursors of invasive carcinoma. A significant difference with classical LCIS (CLCIS) is the distribution at the breast level, since, if CLCIS tends to present in a multifocal manner, the two variants FLCIS and PLCIS almost always have a unifocal pattern. From a genetic point of view, the two variants differ from CLCIS for a greater genetic instability, as a higher rate of alteration of genes coding for tumor suppressors and proteins involved in the regulation of cell growth and replication is documented. Although they express ER and PR they also are HER2 positive more often than CLCIS.

Paget's Disease of the Nipple

- Adenocarcinoma cells occur “locally” in the nipple epidermis, most often due to a cancer in the deeper layers of breast tissue but not necessariily.
- Rare. However, the clinical aspect (--> “nipple eczema”) must be known!

Invasive Breast Cancer – many types

Invasive ductal cancer not otherwise specified (NOS)

- 65 to 80 % of all invasive breast carcinoma
- Tends to be unifocal; additionally, often also peritumoral DCIS
- Easily detectable on mammography
- Originates in the lactiferous ducts: “milk duct cancer”

Invasive Lobular Cancer

- Only 5 to 15 % of all invasive breast carcinomas
- Frequently multifocal, multicentric, bilateral
- Poorly visible on mammogram and ultrasound
- Originates in the glandular acini at the end branches of the milk duct system in the lobules

Inflammatory Breast Carcinoma

- An own entity according to the WHO. This diagnosis is made clinically.
- Poor prognosis if chemotherapy achieves no complete remission.

Invasive Paget's Disease (Paget Carcinoma) of the Nipple

- Rare; however, the clinical aspect (“nipple eczema”) must be known!
- Adenocarcinoma cells in the nipple epidermis, with a subjacent breast cancer or a ductal carcinoma in situ

Particular histologies

- Tubular carcinoma: ER/PR-positive and well differentiated. Best prognosis of all breast carcinomas
- Invasive cribriform carcinoma: Usually ER/PR-positive, favorable prognosis
- Mucinous carcinoma: Usually receptor-positive and well differentiated
- Medullary carcinoma: Axillary metastases are rare despite lack of hormone receptors (triple negative); frequent in women with BRCA1 gene mutations, good prognosis because of numerous tumor-infiltrating lymphocytes
- Papillary, adenoid cystic, secretory, and apocrine carcinomas

Therapy based classification

In practice, invasive ductal and lobular breast cancers are classified based on their therapeutic response (estrogen receptor ER, progesterone receptor PR, and HER2 receptor) into the following categories (figure 1):

- Hormone-sensitive breast cancer (ER positive, incidence: 65 to 75 %)
- HER2-positive breast cancer (can be either ER positive or negative, incidence: 15 to 20 %)
- Triple-negative breast cancer (ER neg, PR neg, HER2 neg, incidence: 15 %)

Hormone sensitive breast cancer:

- Recurrence is rarer than for triple-negative or HER2-positive breast carcinomas.
- However, recurrences occur with the same frequency even 5 to 10 years after diagnosis.

HER2- positive breast cancer:

- Overexpression of the HER2 gene causing overexpression of the HER2 receptor on the cell surface.
- These tumors have a higher metastatic potential than HER2-negative tumors, and they often metastasize to the brain.
- Nevertheless, monoclonal antibodies against HER2 (trastuzumab, Herceptin® and pertuzumab, Perjeta®) offer an immunological therapeutic approach that significantly improves the prognosis, whereby the prognosis equals that of HER2-negative tumors.

Triple negative breast cancer:

- Their gene expression profile often corresponds to the basal cell type, which is not differentiated and has a high risk of metastasis.
- They metastasize to the brain and lungs more frequently than hormone-sensitive carcinomas, but less often to the bones and liver.
- Recurrences usually occur within the first 3 to 5 years after diagnosis, very rarely after 5 years.
- Caution: Medullary or apocrine breast carcinomas are usually triple negative, but their prognosis is significantly better than that of triple-negative invasive ductal or lobular carcinomas.

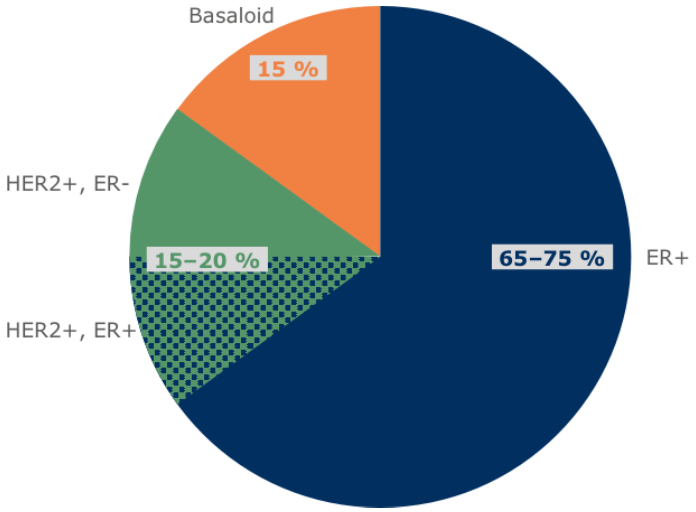

Figure 1: Classification of breast cancer as hormone-sensitive (estrogen receptor positive: ER+), HER2 positive (can be either ER positive or negative), and basaloid carcinomas.

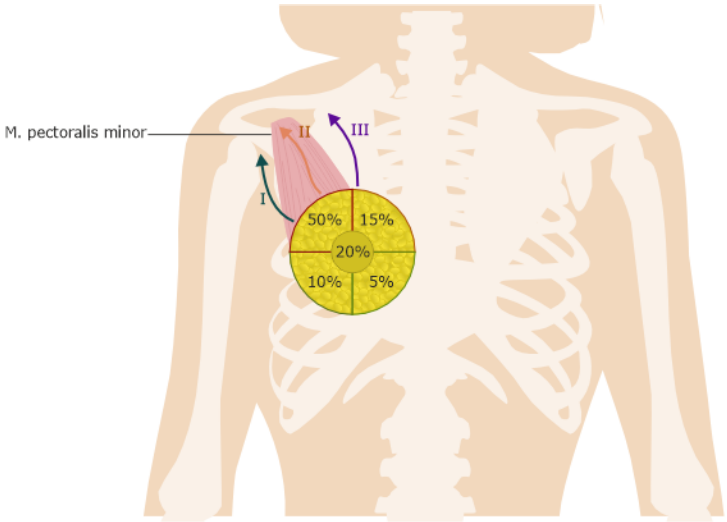

Figure 2: Quadrant localization and lymphatic drainage channels of breast cancer.

Breast cancer quadrant localization (4+1; Figure 2):

- Most breast cancers (50 %) are located in the upper outer quadrant of the breast, since this quadrant contains the largest proportion of gland parenchyma.
- Central carcinomas (e.g., breast cancers located behind the nipple) are the second most common location: 20 %
- Upper inner quadrant: 15 %
- Lower outer quadrant: 10 %
- Lower inner quadrant: 5 %

Lymphogenic metastasis

The axillary lymph node cluster is affected first, and most frequently. There are three anatomic groups or levels of axillary lymph nodes (according to Berg): Level I to Level III (figure 2).

Topographical orientation: Level II is located posterior to the pectoralis minor muscle; Level I is lateral, and Level III is medial, up to the clavicula.

Large lymph channels lead from the breast to the Level I lymph nodes, others lead to the retrosternal lymph node chain ("internal mammary chain"), and rarely directly to the Level II (or III) axillary lymph nodes or infraclavicularly.
Moreover, cutaneous lymph channels lead to the skin areas of the contralateral breast. The lymphogenic direction of metastasis tends to depend on the quadrant within which the respective breast cancer is located, although it is not necessarily correlated with it!

Breast carcinoma with multiple tumor sites

Multifocal breast cancer

- Not rare
- More than one focus of cancer is present in the breast and all foci are in immediate vicinity and in the same quadrant
- Distance between the foci must be < 4 cm

Multicentric breast cancer

- Rare
- More than one focus of cancer is present in the breast, and they are in different quadrants or at least 4 cm apart

Primary bilateral breast cancer (= synchronous bilateral breast carcinomas) only occurs in approximately 1 % of all cases.

If a cancer is diagnosed in the contralateral breast after diagnosing a breast cancer, it is referred to as a metachronous bilateral breast carcinoma. It is also called a contralateral recurrence, even though it is not the same histological tumor variant, and it is typically a secondary carcinoma.

Epidemiology

Breast cancer:

- The incidence is 151/100,000 women per year in Switzerland (National cancer incidence statistics 2023), the age standardized rate is 111,5 per 100’000 person-years.
- Incidence in Germany 122/100’000 --> most frequent type of cancer in women, corresponding to 1/3 of all malignancies in women (incidence in men: see below).
- Approx. 2/3 of patients are aged 60 years or older at diagnosis. The average age of onset is 64 years.
- The risk of disease by age decade is (rounded): 30-39 years: 1:270, 40-49 years: 1:70, 50-59 years: 1:40, 60-69 years: almost 1:25 (figure 3).

- 6485 new cases in Switzerland per year and approximately 1’377 deaths per year.
- The lifetime risk of breast cancer is 12,7% (13 in 100, i.e. one woman in eight); the risk of death from breast cancer is 3,6% for women.
- 70’000 new cases and 17’748 deaths per year in Germany --> most frequent cause of death related to cancer in women.
- In the U.S., breast carcinoma is the second most frequent cause of death related to cancer after lung carcinoma.

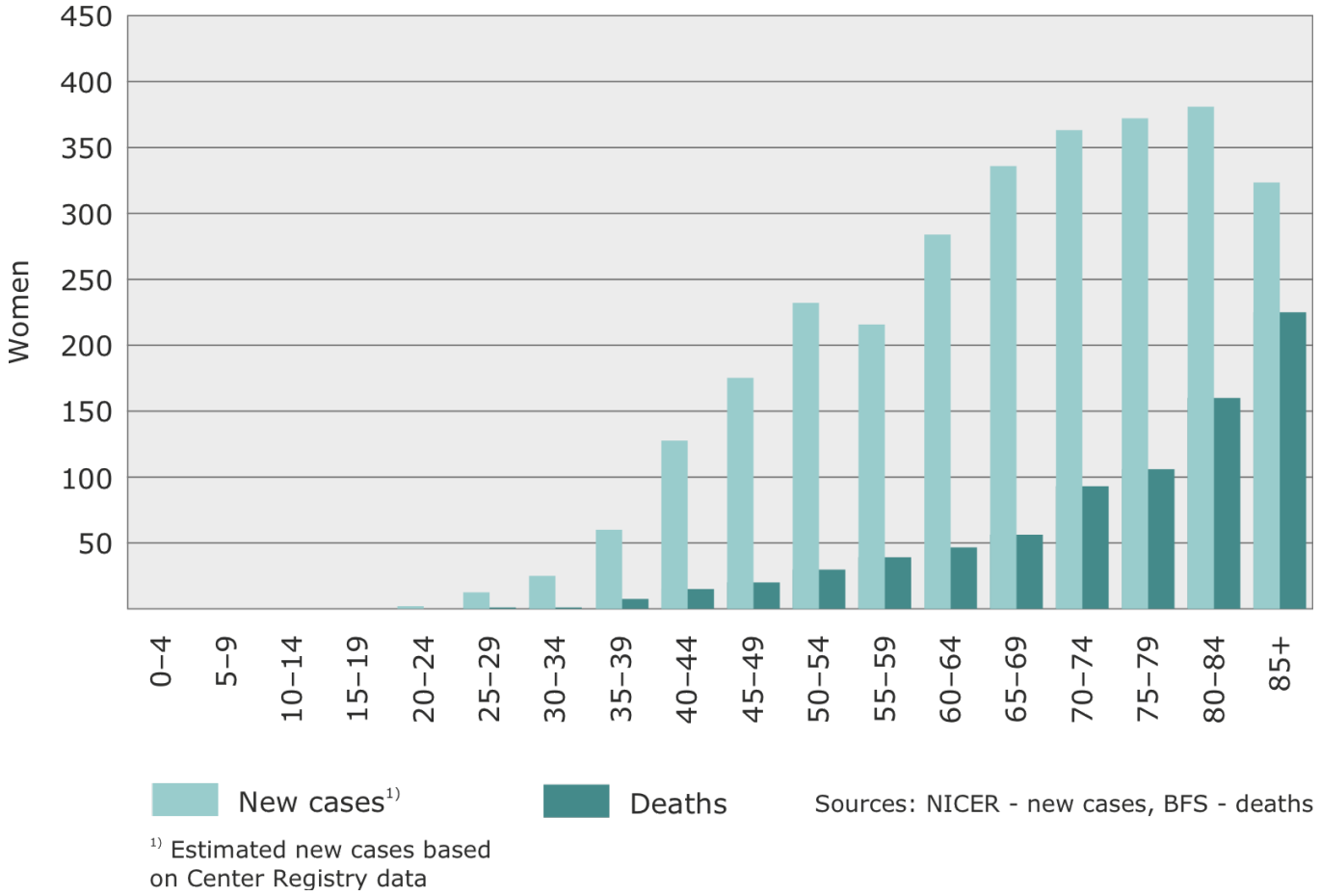

Figure 3: Breast cancer by age, 2016 to 2020 (age-specific rate, per 100’000 population).

One in eight women will suffer breast cancer in her lifetime. One third of all breast cancers occur before the age of 60.

The previously observed North-South disparity in Europe, with a higher frequency and probability of disease in northern European countries, is no longer observed. For example, the incidence in Italy is higher than in Switzerland or Norway.

The incidence rate in Switzerland was on the rise until 2002, yet it has remained constant since (figure 4). Currently, a slightly decreasing trend in the U.S. has been observed for the first time.

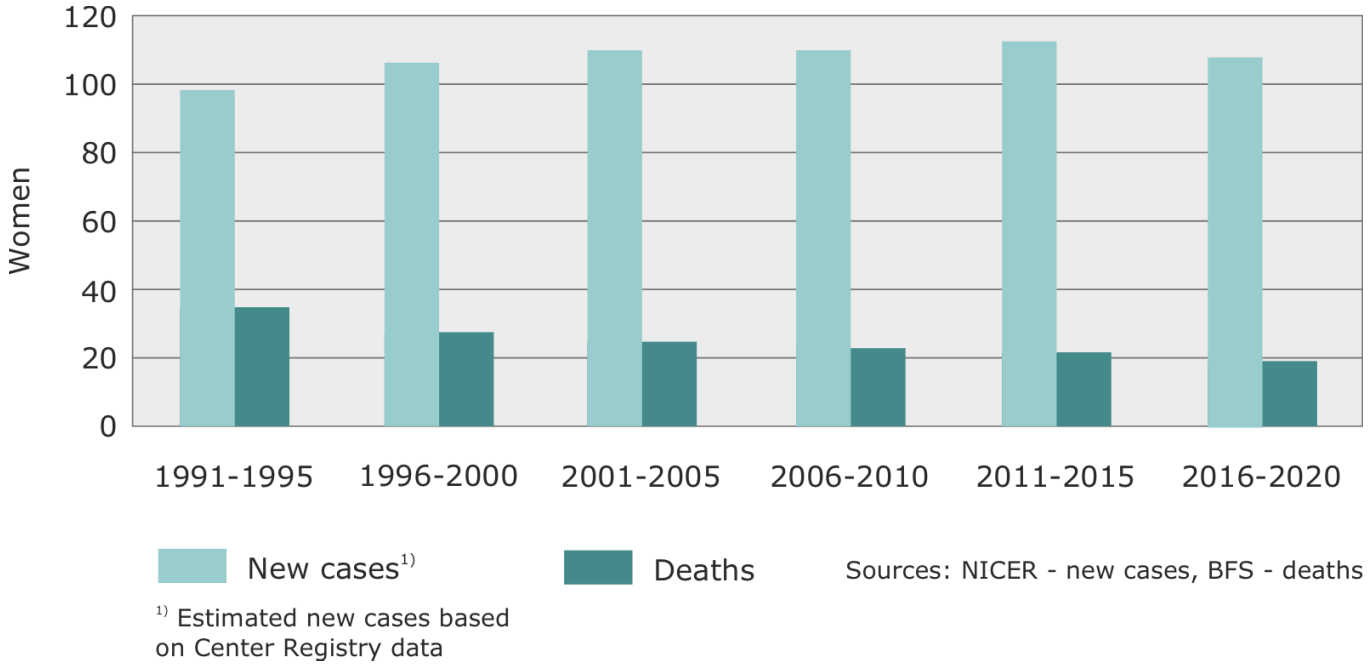

Figure 4: Breast cancer development over time (rate per 100’000 inhabitants, European standard).

Mortality is declining in many Western countries, including Switzerland. This is probably due to more efficient prevention (improved early diagnosis) and multimodal therapies.

Risk factors

Main risk factors

- Age: The main risk factor (see above)!
- Female sex: Less than 1 % of all breast cancers occur in men (frequently in “breast cancer families”; 52 cases/year in Switzerland).
- Positive family history: About 5 to 10 % of all breast cancers are hereditary, i.e., a hereditary genetic mutation can be traced to the father or mother. If the BRCA1 or BRCA2 genetic mutation is present, the risk of suffering breast carcinoma by the age of 80 is 67 % and 66 %, respectively.
- History of breast cancer: 5-times higher risk of having a second de novo breast cancer.
- History of radiation therapy to the mammary glands at an early age: Women who received radiation therapy for lymphoma to part of the mammary glands before the age of 40 have a 5,6-times higher risk of breast cancer. Women who received radiation for Hodgkin's disease at the age of 25 years have a risk of breast carcinoma of 25 to 30 % at the age of 55.
- Positive family history: In the case of PALB2 mutation, the lifetime risk of breast cancer is 45 %, for CDH1 mutation it is 53 %, for CHEK2 mutation, 22 %, and for Peutz-Jeghers syndrome (STK11 mutation), 50 %.

Hereditary breast cancer

The genealogy of approximately 5 to 10 % of all breast cancer patients shows a genetic origin of the disease.

A genetic mutation can only be established in 1/3 of familial breast cancer cases (next-generation sequencing of the 11 breast cancer genes, recommended by the National Comprehensive Cancer Network Guideline).

2 % of all women carry a genetic mutation that promotes breast cancer.

7,8% of all patients with breast carcinoma carry a relevant germline mutation in one of the 11 known genes that promote breast cancer. [P13]

A mutation of the BCRA-1 gene directly contributes to 3,2 % of all breast carcinomas, and a BRCA2 mutation (a DNA repair gene) directly contributes to another 3,1 %. Breast cancer occurs at a younger age in these women and is often triple negative. For instance, the risk of suffering from breast cancer at the age of 35 to 50 years is up to 3 % annually for these gene carriers–compared with 0,1 % for the normal population! The corresponding lifetime risk of breast carcinoma for these gene carriers is 67 %. Note: These mutation carriers also have an increased risk of ovarian cancer (45 %).

The BRCA1 gene is located on chromosome 17; the BRCA2 gene on chromosome 13. Both are tumor suppressor genes that contribute to the stability and repair of DNA. Mutations of these genes are inherited autosomal dominantly. If the second allele also loses its function, the respective protein can no longer be produced correctly, leading to DNA repair disruption and malignant degeneration.

If a woman with a BRCA mutation was treated for breast cancer, the risk of contralateral breast carcinoma increases as well. The cumulative 10-year risk for non-mutation carriers is 6 %, for BRCA2 mutation carriers, 11 %, and for BRCA1 mutation carriers, 20 %.

Less than 4 % of breast carcinomas are caused by rare genetic syndromes such as:

- CHEK2 mutation (1,6 %)
- PALB2 (1 %)
- ATM mutation (0,7 %, ataxia telangiectasia)
- TP53 mutation (0,3 %)
- PTEN gene mutation (0,06 %, Cowden syndrome)
- CDH1 (0,14 %, hereditary diffuse gastric cancer syndrome)
- STK11 (0,03 %, Peutz-Jeghers syndrome)
  [P13]

There are various models for determining the risk of breast cancer based on personal history without genetic testing. None of the methods can integrate all modifying factors concurrently. The Gail model takes into account menarche, menopause, age at first delivery, and number of relatives with breast carcinoma, but not relatives with ovarian carcinoma. As a result, possible heredity tends to be underestimated. The Claus-Model can only include two affected relatives in the risk calculation and does not consider ovarian carcinoma either [P11, P12].

For an estimate of familial risk and indication for genetic counseling, see Guidelines for the management of women at increased familial risk of breast cancer [B02]. [B02] [P11][P12]

Empirical mutation detection rates as a function of family history

- ≥ 3 breast carcinomas, 2 of them premenopausal: 30,7 %
- ≥ 3 breast carcinomas: 22,4 %
- 2 premenopausal breast carcinomas: 19,3 %
- 2 breast carcinomas, 1 of them premenopausal: 9,2 %
- ≥ 1 breast carcinoma and > 1 ovarian carcinoma: 48,4 %
- ≥ 2 ovarian carcinomas: 45 %
- 1 breast carcinoma ≤ 36 years of age: 10,1 %
- 1 bilateral breast carcinoma, premenopausal: 24,8 %
- 1 breast carcinoma in a male and 1 breast or ovarian carcinoma: 42,1 %

Figure 5 shows the genetic mutations associated with an increased risk of breast carcinoma.

		AR	RR		Relative risk of other cancer			
High-risk gene	**Syndrome**	**BC (80y)**	**BC**	**Ovarian**	**Colon**	**Stomach**	**Renal**	**Other**
BRCA1	HBOC	67 % male: 1 %	4,0–7	45 %				Melanoma cumulative 2,5 %*
BRCA2	HBOC	66 % male: 7 %	4,0–7	12 %				Melanoma cumulative 2,3 %*
BRIP1				↑				
CDH1	HDGC	53 %	5,9–7,3					
PALB2	-	45 %	1,4–3,9					
PTEN	Cowden		2,0–5,0					
TP53	Li Fraumeni		4,3–9,3					
MLH1	HNPCC	-	0–4,2 +	↑	52–69 %	↑		Endo-
MSH2	HNPCC	-	0,8–4,5 +	↑	52–69 %	↑		metrium
MSH6	HNPCC	-	0–11,7 +	↑	52–69 %	↑		70 %
PMS2	HNPCC	-	-	↑	52–69 %	↑		
CHEK2	0,8 %	22 %	2,1	-	-	5,8	3,6	Sarkome 3,5
STK11	Peuth-Jeghers	50 %		↑		↑		

Figure 5: Genetic mutations associated with an increased risk of breast carcinoma.
(AR = absolute risk in %, RR = relative risk compared to the average population. BC = breast cancer, HBOC = hereditary breast and ovarian carcinoma, HNPCC = hereditary non-polyposis colon carcinoma
* Results from a study published in the journal “Hereditary Cancer in Clinical Practice” showed that the risk of non-melanoma skin cancer in women who carry BRCA1 or BRCA2 mutations is similar to that of noncarrier women. However, the risk of melanoma appears to be slightly elevated. In the study, the cumulative risk of melanoma was 2.5% for BRCA1 carriers and 2,3 % for BRCA2 carriers, compared with 1,5 % for women in the general population.

If no genetic testing is available, the individual risk may be estimated by family history (table 2).

Table 2: Family history and lifetime risk of breast cancer

Breast cancer (MC)	Lifetime risk	Risk group	Annual mammography, 40-50 years	Indication for genetic counseling
1 relative < 40 years	1:6	Moderate risk	Yes	No, except Ashkenazi
2 relatives < 50 years and > 40 years	1:4-5	Moderate risk	Yes	Yes

Breast cancer (MC)	Lifetime risk	Risk group	Annual mammography, 40-50 years	Indication for genetic counseling
2 relatives < 60 years and > 50 years	1:5-6	Moderate risk	Yes	No, except Ashkenazi
3 relatives < 60 years	1:4	Moderate risk	Yes	No, except Ashkenazi
1 relative with bilateral MC	1:3-6	Moderate risk, high risk if < 40 years	Yes	No, except Ashkenazi or < 50 years
2 relatives < 40 years	1:3-4	High risk	Yes	Yes
3 relatives < 50 years	1:3	High risk	Yes	Yes
4 relatives of any age	1:2 to 1:3	High risk	Yes	Yes
1 OC any age + 1 MC < 50 years	1:3-6	Moderate to high	Yes	Yes
> 1 OC	1:3	High	Yes	Yes

Hormone dependent risk factors (RR = relative risk)

Table 3: Hormone dependent breast cancer risk factors

Increasing endogenous exposure to estrogen	early menarche late menopause late or no deliveries; short periods of breastfeeding	RR < 2 RR < 2 RR for nullipara: 1,7

Increasing endogenous exposure to estrogen	long-term combined estrogen replacement therapy after menopause oral contraceptives before first delivery	RR 1,23 RR 1,1
Proliferating lesions of the mammary gland	atypical ductal hyperplasia atypical lobular hyperplasia	10-year risk: 17 % 10-year risk: 20 %
Mammography density of the mammary gland	very high density (ACR 4, see below)	RR > 4
Very high mammography density with hyperplasia but without atypia	very high density (ACR 4)	RR 2
Very high mammography density plus atypical hyperplasia	Very high density (ACR 4)	RR 5,3

Hence, the following reduce risk: multiparity; breastfeeding (RR 0,7).

Empirically, “mammographic density” (figure 6) is a risk factor. The risk of breast cancer increases with increasing mammographic density. According to the American College of Radiology, there are 4 levels of mammographic density. However, this is not an independent risk factor, but an expression of various factors, such as cumulative exposure to estrogen of the breast gland.[B03][B07]

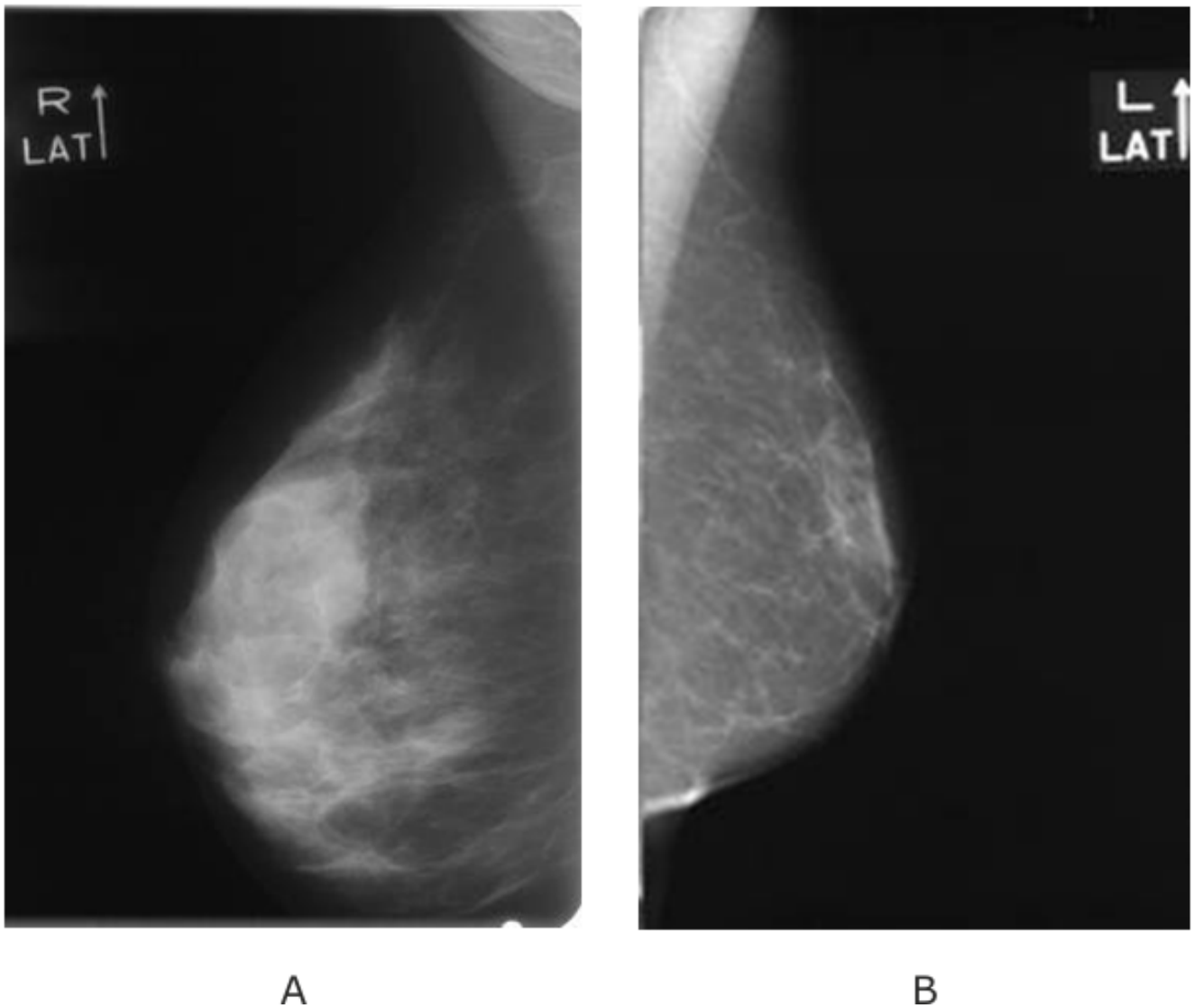

Figure 6: High mammographic density of the breast parenchyma (left) indicates 4 to 6 times higher risk of developing breast carcinoma than low density (right).

Risk of breast cancer and hormone replacement therapy

Analysis of epidemiological studies requires a distinction to be made between patients currently undergoing and patients who previously underwent hormone replacement therapy.

- After discontinuation of hormone replacement therapy, the risk of breast cancer decreases to the risk for non-users within 1 to 4 years.
- Estrogen monotherapy presented a decreased or minimally increased risk of breast carcinoma (plus 1 % within 20 years).
- In contrast, combined estrogen/progestogen hormone replacement therapy presented an increased risk of breast cancer after more than 5 years of therapy (RR 1,24).

The evidence was confirmed in several studies:

- After 5 years of combined estrogen/progestogen hormone replacement therapy, the incidence of breast carcinoma increases by approximately 8 additional cases in 10'000 women per year compared to women without hormone replacement therapy.
- Estrogen-only therapy does not show any increased risk until 7 years of treatment. Then, observational studies show a slightly increased risk.
- Women who developed breast cancer during hormone replacement therapy have a lower mortality than women with breast carcinoma and without previous hormone replacement therapy.
- Combined hormone replacement therapy leads to increased mammographic breast density (as a surrogate for the risk of breast carcinoma) in approximately 50 % of treated women. Estrogen mono-substitution has a lower effect. This effect on mammographic breast density is dose-dependent and is no more than 5 to 10 %.
  [P02][P03][P05]

For breast cancer survivors, hot flashes are a significant problem which affects quality of life. Since tibolone (Livial®) is supposed to stimulate glandular breast tissue less than estrogen substitution, tibolone was tested in a randomized study in patients with a previous breast carcinoma. Unfortunately, the women treated with tibolone had a significantly higher risk of recurrence (RR 1,45), so this synthetic estrogen is also not recommended. [P01]

Other, minor risk Factors

- Obesity (only for postmenopausal breast carcinoma): BMI > 30: relative risk (RR) 2
- Consumption of alcohol (linear risk increase): > 3 units per day: RR 1,4
- Height (taller women have a higher risk): low risk: approx. RR 1,2
- Long-term night shift work: RR 1,5
- Lack of physical activity: Physical activity lowers the risk of premenopausal and postmenopausal breast carcinoma. However, the difference for premenopausal women in the incidence of breast cancer was only 6% when the quartile of the highest physical activity was compared to the quartile with the lowest. For HER2 positive breast cancer the risk reduction was higher, namely 43 %
- Smoking: Highest risk for smoking before the first delivery: RR 1,3
  [B05][P04][P09]

Vitamin D3 deficiency increases the risk of breast cancer recurrence after treatment, but data are unclear concerning primary breast carcinoma.

Literature

Nr.	Title	Author	Journal	Volume	Pages	Year
[B01]	WHO Classification of Tumours of the Breast	Lakhani, S.R., Ellis. I.O., Schnitt, S.J., Tan, P.H., van de Vijver, M.J	IARC WHO Classification of Tumours	4		2012
[B02]	Clinical Practice Guidelines in Oncology	NCCN				2016
[B03]	Mammographic breast density as an intermediate phenotype for breast cancer	Boyd NF, Rommens JM, Vogt K, et al	Lancet Oncol	6	798-808	2005

Nr.	Title	Author	Journal	Volume	Pages	Year
[B04]	Breast cancer epidemiology according to recognized breast cancer risk factors in the Prostate, Lung, Colorectal and Ovarian (PLCO) Cancer Screening Trial Cohort	Lancey JV Jr., Kreimer AR, Buys SS, Marcus PM, Chang SC, Leitzmann MF, Hoover Rn, Prorok PC, Berg CD, Hartge P, Prostate, Lung, Colorectal and Ovarian (PLCO) Cancer Screening Trial Project Team	BMC Cancer	9	84	2009
[B05]	Nightshift work and risk of breast cancer and other cancers – a critical review of the epidemiologic evidence	Kolstad AK	Scand J Work Environ Health	34	5-22	2008
[B06]	Secondary breast cancer in Hodgkin's lymphoma survivors	Crump M	J Clin Oncol	27	4229-4231	2009

Nr.	Title	Author	Journal	Volume	Pages	Year
[B07]	Association of active and passive smoking with risk of breast cancer among postmenopausal women: a prospective cohort study	Luo J, Margolis KL, Wactawski-Wende J, Horn K, Messina C, Stefanick ML, Tindle HA, Tong E, Rohan TE	BMJ	342	1016	2011
[B08]	Risk-Reducing Surgery in Hereditary Breast and Ovarian Cancer	Hartmann LC, Lindor NM	N ENGL J MED	374;5	455	2016

Nr.	Title	Author	Journal	Volume	Pages	Year
[P01]	Safety and efficacy of tibolone in breast-cancer patients with vasomotor symptoms: a double-blind, randomised non-inferiority trial	Kennemans P, Bundred NJ, Foidart JM, Kubista E et al	Lancet Oncol	10	135-146	2009
[P02]	Prediagnostic use of hormone therapy and mortality after breast cancer	Newcomb PA, Egan KM, Trentham-Dietz A, et al	Cancer Epidemiol Biomarkers Prev	17	864-871	2008

Nr.	Title	Author	Journal	Volume	Pages	Year
[P03]	Unopposed estrogen therapy and the risk of invasive breast cancer	Chen WY, Manson JE, Hankinson SE, et al	Archi Intern Med	166	1027-1032	2006
[P04]	Exercise and breast cancer prevention	Bernstein L	Curr Oncol Rep	11	490-496	2009
[P05]	Health risks and benefits 3 years after stopping randomized treatment with estrogens and progestins	Heiss G, Wallace R, Anderson GL, Agaraki A, Berensford SA, Brzyski R, Chlebovski RT, Gass M, LaCroix A, Manson JE, Prentice RL, Rossouw J, Stefanick ML	JAMA	209	1036-1045	2008
[P06]	Breast cancer subtypes and the risk of local and regional relapse	Voduc KD, Cheang MC, Tyldesley S, Gelmon K, Nielsen TO, Kennecke H	J Clin Oncol	28	1684-1691	2010

Nr.	Title	Author	Journal	Volume	Pages	Year
[P07]	Benign Breast Disease, Mammographic Breast Density, and the Risk of Breast Cancer	Tice JA, O'Meara ES, Weaver DL, Vachon C, Ballard-Barbash R, Kerlikowske K	J Natl Cancer Inst	105(14)	1043-1049	2013
[P08]	Active smoking and breast cancer risk: original cohort data and meta-analysis	Gaudet MM, Gapstur SM, Sun J, Diver WR, Hannan LM, Thun MJ	J Natl Cancer Inst	105	515-525	2013
[P09]	Smoking and Mortality — Beyond Established Causes	Carter BD et al	N Engl J Med	372	631	2015
[P10]	Prospective identification of tumorigenic breast cancer cells	Al-Hajj M, Wicha MS, Benito-Hernandez A, Morrison SJ, Clarke MF	Proc Natl Acad Sci USA	100	3983–3988	2003
[P11]	Projecting individualized absolute invasive breast cancer risk in African American women	Gail MH, Costantino JP, Pee D, et al	J Natl Cancer Inst 99	23	1782-1792	2007

Nr.	Title	Author	Journal	Volume	Pages	Year
[P12]	Application of breast cancer risk prediction models in clinical practice	Domchek SM, Eisen A, Calzone K, et al	J Clin Oncol 21	4	593-601	2003
[P13]	J Clin Oncol	Kurian AW, Ward KC, Howlader N, et al		37	1305-1315	2019